

Data Analysis & AI Fundamentals

Orientation & Introductory Session

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June 15, 2026

Welcome to the Program

Welcome!

Welcome to the **Data Analysis & AI Fundamentals Scholarship Program.**

- A **6-week intensive** certification course.
- Designed to empower you with industry-relevant skills.
- Nurturing analytical thinking and inspiring innovation.
- Preparing you for opportunities in the growing data ecosystem.

Victor Wandera Lumumba, MSc.

- **Education:** MSc in Applied Statistics & BSc in Economics and Statistics (Chuka University).
- **Expertise:** Machine Learning, Health Data Modeling, Survival Analysis, and Explainable AI (XAI).
- **Current Roles:**
 - Part-Time Assistant Lecturer, Chuka University.
 - Data Analyst & Trainer at CDAM, Chuka University.
 - Data Analysis and Machine Learning Trainer-Mediareast Training College

- **Research Focus:** Building interpretable ML models for risk prediction in health and finance.
- **Publications:** Multiple papers in computational biology, artificial intelligence, and statistics.
- **Applied Projects:** CardioPredict AI, DiabetesAI, AI Mental Distress Screening App.
- **Web:** beyonddataanalytics.online

Session Agenda

- 1 Program Overview & Expectations
- 2 Introduction to Data Analysis
- 3 Types of Data & Common Formats
- 4 Data Collection Methods
- 5 Data Quality & Preprocessing
- 6 Tools & Next Steps

Program Overview

Course Details

- **Course:** Data Analysis & AI Fundamentals
- **Duration:** 6 Weeks Intensive Certification
- **Mode:** Physical and Online (Flexible)

Empowering youth and professionals with industry-relevant data analysis and AI skills.

Program Objectives

- Nurture analytical thinking.
- Inspire innovation in the data ecosystem.
- Equip participants with industry-standard tools.
- Prepare learners for opportunities in Data & AI.

Who Can Join This Program?

- Students & recent graduates
- Early-career professionals
- Entrepreneurs & startup founders
- Researchers
- Non-tech professionals
- IT & CS learners
- Job seekers
- AI enthusiasts!

Course Outline Overview

- ① **Foundations of Data Analysis** (Weeks 1-2)
- ② **Data Visualization & Essential Analytics** (Weeks 3-4)
- ③ **Machine Learning & AI Fundamentals** (Week 5)
- ④ **AI Applications, Ethics & Capstone Project** (Week 6)

Module 1: Foundations of Data Analysis

- Understand what data analysis is and why it matters.
- Learn types of data and common data formats.
- Introduction to Python, Pandas, and NumPy.
- Basic descriptive statistics.

Module 2: Data Visualization & Analytics

- Data visualization principles (Matplotlib, Seaborn).
- Correlation, distributions, and trends.
- Data cleaning and preprocessing.
- Feature engineering basics.

Module 3: Machine Learning & AI

- Core AI/ML concepts and pipelines.
- Supervised vs. unsupervised learning.
- Regression & classification models.
- Model evaluation (Accuracy, Precision, Recall, F1).

Module 4: AI Applications & Capstone

- AI use cases in finance, health, marketing.
- Responsible AI, bias, fairness, and privacy.
- Introduction to Generative AI (ChatGPT, LLMs).
- End-to-end final project and presentation.

Let's explore the core concepts of data, analysis, and quality.

What is Data?

Definition

Raw facts, figures, and statistics collected for reference or analysis.

- **Characteristics:** Can be quantitative (numbers) or qualitative (descriptions).
- **Role:** The foundational building block of any analytical or AI model.
- **Context:** Data alone has no meaning; it requires analysis to become *information*.

What is Data Analysis?

- The process of inspecting, cleansing, transforming, and modeling data.
- **Goal:** To discover useful information, inform conclusions, and support decision-making.
- Bridges the gap between raw data and actionable insights.

The Data Analysis Lifecycle

- ➊ **Ask:** Define the problem or question.
- ➋ **Collect:** Gather the necessary data.
- ➌ **Clean:** Handle missing values and outliers.
- ➍ **Analyze:** Apply statistical or ML techniques.
- ➎ **Visualize:** Communicate findings effectively.
- ➏ **Act:** Make data-driven decisions.

Why Data Analysis Matters

- **Decision Making:** Moves organizations from intuition-based to evidence-based decisions.
- **Efficiency:** Identifies bottlenecks and optimizes operations.
- **Personalization:** Enables tailored experiences (e.g., Netflix, Amazon).
- **Risk Management:** Predicts failures, fraud, or health crises.

Real-World Applications

- **Healthcare:** Predicting disease outbreaks, patient readmission risks.
- **Finance:** Credit scoring, algorithmic trading, fraud detection.
- **Marketing:** Customer segmentation, churn prediction.
- **Agriculture:** Crop yield forecasting using climate data.

Not all data is created equal.

Understanding data types dictates how we store, process, and analyze it.

Broad Categories of Data

- **Quantitative:** Numerical (Discrete vs. Continuous).
- **Qualitative (Categorical):** Nominal vs. Ordinal.
- **Structural Format:** Structured, Unstructured, and Semi-Structured.

Definition

Highly organized, typically in tabular format with rows and columns.

- **Characteristics:** Easily searchable, fits neatly into Relational Databases (SQL).
- **Examples:** Excel spreadsheets, CSV files, SQL databases, sensor logs.

Definition

Data without a predefined data model or organization.

- **Characteristics:** Heavy, complex, difficult to search and analyze using traditional tools.
- **Examples:** Emails, social media posts, images, audio files, videos, PDF documents.

Definition

Does not conform to a strict tabular structure but contains tags or markers.

- **Characteristics:** Flexible, self-describing, bridges structured and unstructured.
- **Examples:** JSON, XML, HTML, NoSQL databases (e.g., MongoDB).

Structured vs. Unstructured: Comparison

Feature	Structured	Unstructured
Format	Tabular (Rows/Cols)	No predefined model
Storage	RDBMS (SQL)	Data Lakes / NoSQL
Analysis	BI / Statistics	NLP / Computer Vision
Searchability	Highly searchable	Complex to search

Common Data Formats: Tabular

- **CSV (Comma Separated Values):** Simple, lightweight, widely used.
- **Excel (.xlsx):** Supports multiple sheets, formatting, and basic formulas.
- **Parquet / Feather:** Columnar storage, highly optimized for big data and Python/R.

Common Data Formats: Hierarchical

- **JSON (JavaScript Object Notation):** Key-value pairs, standard for web APIs.
- **XML (eXtensible Markup Language):** Uses tags, heavily used in enterprise systems.
- **YAML:** Human-readable, often used for configuration files.

Common Data Formats: Multimedia/Text

- **Images:** JPEG, PNG, TIFF (used in Computer Vision).
- **Audio:** WAV, MP3 (used in Speech Recognition).
- **Text:** TXT, PDF, DOCX (used in Natural Language Processing).

How do we get the data we need for analysis?

What is Data Collection?

- The systematic process of gathering and measuring information on targeted variables.
- Enables us to answer stated questions, test hypotheses, and evaluate outcomes.
- **Crucial step:** Poor collection invalidates even the best ML models.

Primary vs. Secondary Data

- **Primary Data:** Collected first-hand for a specific research problem (e.g., surveys, experiments, IoT sensors).
- **Secondary Data:** Already collected by someone else for a different purpose (e.g., government census, Kaggle datasets, published research).

Modern Data Collection Methods

- **APIs (Application Programming Interfaces):** Pulling live data from web services (e.g., Twitter API, Weather APIs).
- **Web Scraping:** Extracting data from websites using Python (BeautifulSoup, Scrapy).
- **IoT & Sensors:** Continuous streaming data from devices (e.g., wearables, smart meters).

Collecting data is only half the battle.

Is the data actually good?

Garbage In, Garbage Out (GIGO)

The Golden Rule of Data Science

Flawed, biased, or nonsense input data produces equally flawed output.

- In AI/ML: If you train a model on inaccurate data, predictions will fail.
- **Fact:** Data cleaning and preprocessing often take **60% to 80%** of a data scientist's time!
- High-quality data is more important than complex algorithms.

Key Dimensions of Data Quality

- ① **Accuracy:** Does the data reflect reality?
- ② **Completeness:** Are there missing values or records?
- ③ **Consistency:** Is data uniform across different datasets?
- ④ **Timeliness:** Is the data up-to-date?
- ⑤ **Validity:** Does the data conform to defined syntax/format rules?

Common Data Quality Issues

- **Missing Values:** Blank cells, NaNs, or nulls.
- **Duplicates:** Exact or near-exact repeated records.
- **Outliers:** Extreme values that deviate from other observations.
- **Inconsistent Formatting:** Dates written as "DD/MM/YYYY" and "Month DD, YYYY" in the same column.
- **Typos/Errors:** "Niarobi" instead of "Nairobi".

Strategies for Data Cleaning

- **Handling Missing Data:** Imputation (mean, median, ML-based) or dropping rows/columns.
- **Removing Duplicates:** Identifying and deleting redundant entries.
- **Outlier Treatment:** Capping, transforming, or removing extreme anomalies.
- **Standardization:** Converting text to lowercase, standardizing date formats.

Tools We Will Cover in This Course

- **Python:** The core programming language for Data Science.
- **Pandas & NumPy:** For data manipulation and numerical computing.
- **Matplotlib & Seaborn:** For creating compelling data visualizations.
- **Scikit-Learn:** For building and evaluating machine learning models.
- **Jupyter Notebooks:** Our interactive coding and documentation environment.

Expectations & Next Steps

- **Commitment:** Attend all sessions and complete hands-on assignments.
- **Practice:** Coding is a skill; practice daily.
- **Collaboration:** Work together, ask questions, and share insights.
- **Capstone:** Prepare to build an end-to-end project by Week 6!

Questions?

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